



RAMA REAL-ESTATE

Dubai market + product technical blueprint

A trust-first real-estate platform for buying, renting and investing with verifiable evidence, life-fit intelligence and immersive property tours.

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CONFIDENTIAL STRATEGY DOCUMENT

From market signal to buildable system.

RAMA is positioned as an evidence layer across the property lifecycle—not a bigger catalogue of listings.

The document separates verified facts from strategy. Superscript numbers connect claims to the source register. Statements marked **RAMA synthesis** are product inferences from those facts and from a review of public product surfaces; they are not claims of exhaustive competitor absence.

Scope. Dubai residential real estate, with ready and off-plan ownership plus long-term rental. Commercial, holiday homes, brokerage licensing operations and tokenised ownership are adjacent opportunities, not MVP scope.

This document is a product and technical strategy—not legal, tax, valuation or investment advice. Regulations, eligibility, fees, API access and mortgage products must be revalidated before launch.

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Less persuasion. More proof.

Dubai does not need another inventory-heavy portal. It needs a calm, regulator-aware decision environment that turns fragmented evidence into confident action.

Market reality

Transaction value, rental activity, project delivery and brokerage participation are all expanding. The result is choice—and a heavier verification burden for the customer.

Experience gap

Public market leaders already offer verified listings, transaction data, commute search, floor plans and 360 tours. The next frontier is not another feature; it is evidence joined across the whole decision.

RAMA wedge

A Trust Passport for every property, full-cost and net-return models, inspection-grade tours, and human support designed around first-time, remote and access-conscious customers.

“A property page should behave like a due-diligence room, a viewing appointment and a life simulator—without feeling like any of them.”

01

Verify

Source, timestamp and confidence for every important claim.

02

Understand

Plain-language costs, tenure, risk and neighbourhood trade-offs.

03

Experience

A tour that answers questions—not merely rotates a panorama.

04

Continue

One decision record from discovery through handover and ownership.

A high-velocity market with growing decision load.

AED 761B

2024 real-estate transaction value; 226,000 transactions.¹

AED 252B

Q1 2026 transaction value; +31% year on year.²

1.38M

Registered tenancy contracts in 2025; AED 126.4B value.³

937

Projects under construction in 2025; +25% year on year.³

What the numbers say



Bars are visually normalized within this panel; they do not share a unit or imply a forecast. Figures from DLD.¹²³

RAMA synthesis

Market growth increases the cost of a bad decision. More projects, offices and transactions mean more inventory states, sales narratives, fees, milestones and counterparties to reconcile.

- **Search is abundant;** verified interpretation is scarce.
- **Off-plan is attractive;** delivery evidence is fragmented.
- **Transactions are visible;** net ownership economics remain work.
- **Virtual tours exist;** inspectability and provenance can go deeper.

Research date: 17 July 2026. The market pulse uses the latest official period-specific figures identified during research; annual, rental and quarter snapshots are intentionally not blended into a synthetic forecast.

Dubai's policy direction rewards transparent PropTech.

2033

Real Estate Sector Strategy

Targets include ~AED 73B sector GDP contribution, 33% homeownership, +70% transactions and AED 1T market value. The strategy names data centralisation, AI, sustainability and an integrated customer lifecycle.⁴

2040

Urban Master Plan

Plans for 5.8M residents by 2040, five urban centres, doubled green/recreation areas and 55% of residents within 800m of mass transit.⁵

2029

Metro Blue Line

Scheduled to open 9 September 2029, expanding rail reach into growth areas. RAMA should model "today / committed / scenario" accessibility separately.⁶

2025

First-Time Home Buyer Programme

Open to UAE residents aged 18+ across nationalities and income levels who do not own a Dubai freehold home; offers priority launch access, preferential pricing and tailored mortgages.⁷

Build for "proof"

Use DLD-linked identifiers, source dates, permit verification and confidence labels.

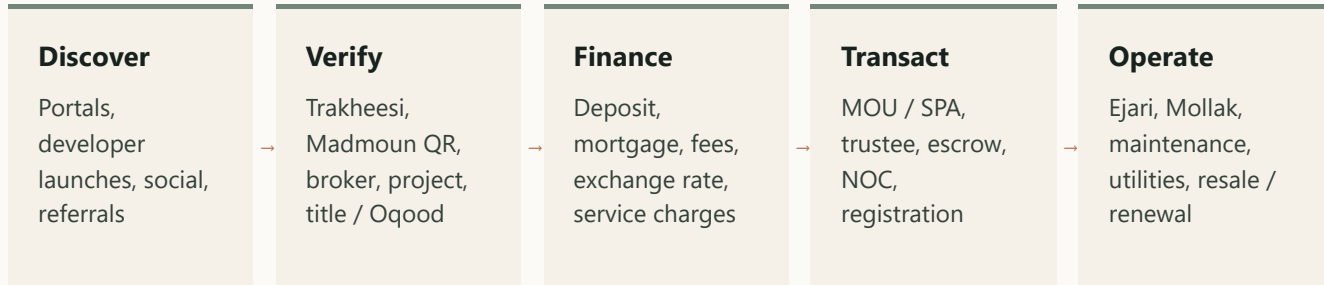
Build for "balance"

Serve renters, owner-occupiers and investors; avoid a luxury-only editorial lens.

Build for "future state"

Keep committed infrastructure distinct from present-day travel and services.

The customer crosses systems. Their decision should not.



Regulatory / service primitives

PRIMITIVE	ROLE IN RAMA
Madmoun / Trakheesi	Advertisement permit and QR authenticity. QR became mandatory for real-estate ads in 2023. ⁸
Dubai REST	Government service surface for title checks, brokers, project status, rental index, Ejari and more. ⁹
Escrow / Oqood	Off-plan payment safeguarding and interim registration context. ¹⁰
Ejari / Mollak	Tenancy registration and jointly owned property / service-charge lifecycle.
DLD API gateway	Potential partner integration for brokers, Mollak, Ejari and Trakheesi—access subject to approval and contract. ¹¹

The RAMA continuity layer

Maintain one canonical **Decision Record** that survives channel changes: shortlist → tour → offer → finance → handover → ownership.

Evidence ledger

Source, retrieval date, scope, confidence, expiry.

Consent ledger

Purpose, channel, lawful basis, withdrawal.

Document room

Role-based access, malware scan, watermark, audit trail.

Task trail

Owner, SLA, handoff and next best action.

Strong discovery tools. Open space after the click.

PUBLICLY VISIBLE CAPABILITY	PROPERTY FINDER	BAYUT	EMAAR	NAKHEEL
DLD transaction exploration	Prominent	Prominent	Not core	Not core
Listing / agent trust signal	SuperAgent + verified listing claims	TruCheck + broker profiles	First-party stock	First-party stock
Mortgage / upfront cost support	Mortgage calculator + quote	Editorial / tools	Project-specific	Partner financing
Commute-aware discovery	Community taxonomy	Two-location commute search	Community narrative	Community narrative
3D plans / 360 tours	Listing dependent	45K+ plans; 360 / 3D	Selected 360 VR tours	Media-led project pages
Post-purchase service	Not core	Not core	Developer channels	App: service, payments, support
Cross-inventory evidence passport	Not prominent	Not prominent	Own stock	Own stock
Inspection-grade tour annotations	Not prominent	Not prominent	Not prominent	Not prominent

Sources: Property Finder homepage, mortgage and SuperAgent pages;¹²¹³ Bayut Transactions, app and TruCheck pages;¹⁴¹⁵ Emaar VR Tours;¹⁶ Nakheel launches, app and FAQ.¹⁷ “Not prominent” means not evident on reviewed public surfaces—not proof the organisation lacks the capability.

Do not compete on breadth alone

Inventory aggregation produces a costly, portal-shaped business. RAMA’s initial supply should be curated and evidence-complete.

Do not call 360° “innovation”

360 and 3D plans are established. Differentiate on measurement, inspectability, provenance, synchronized guidance and decision capture.

Connect lifecycle seams

The defensible asset is the verified decision graph across property, person, evidence, tour, cost and task.

Seven gaps worth owning.

01 · Decision-grade truth

Beyond “verified listing”

Show which attributes are regulator-sourced, broker-attested, captured on site, modelled or unknown. Every badge must answer: verified by whom, when, against what?

02 · Full cost of ownership

Beyond mortgage payment

Combine purchase costs, financing, service charges, cooling, maintenance reserve, vacancy, management and currency scenarios with assumptions users can edit.

03 · Off-plan observability

Beyond launch material

Project timeline, escrow and registration evidence, construction updates, variation log, handover checklist and scenario risk in one chronology.

04 · Life-fit evidence

Beyond amenities

Peak-time commute, shade and walking comfort, stroller / wheelchair route, school run, pet constraints, acoustic context, delivery access and actual view orientation.

05 · Remote buyer operating model

Beyond a WhatsApp thread

Asynchronous tour, time-zone aware SLA, structured questions, secure documents, multilingual summaries and an authorized representative workflow.

06 · Accessibility by evidence

Beyond a generic filter

Door clearances, step-free path, lift dimensions, bathroom turning radius and parking route—captured, timestamped and clearly distinguished from formal certification.

07 · Calm advice

Beyond urgency mechanics

Explain trade-offs and uncertainty; cap urgency patterns; record why a recommendation fits. Nordic restraint becomes a trust mechanism, not decoration.

Opportunity filter

A gap enters the roadmap only when RAMA can source reliable data, explain the method, maintain freshness, and create measurable customer value without implying regulatory authority.

Start with people whose decision work is structurally harder.

First-time resident buyer

"Can I safely move from rent to ownership?"

Friction: deposit and fee shock, mortgage eligibility, jargon, tenure, process sequencing. Expat first-home LTV may reach 80% at ≤AED 5M, subject to lender assessment; maximum mortgage term is 25 years.¹⁸

RAMA promise: readiness plan, programme eligibility checklist, cost waterfall, document checklist, human review before offer.

Remote / non-resident investor

"Show me what I cannot inspect myself."

Friction: time zones, delegated viewing, source trust, FX, financing, ownership operation and handover. A glossy render is least useful where physical inspection is impossible.

RAMA promise: remote evidence pack, guided live tour, annotated defects / unknowns, net-return scenarios, secure representative workflow.

Woman buying independently

"Let me control privacy, safety and advisory style."

Signal: women accounted for 34% of investors and 50,979 transactions valued at AED 118B in DLD's 2024 figures.¹⁹

RAMA promise: privacy-preserving contact, advisor preference, co-browsing without phone exposure, safe viewing logistics, equal evidence—not a cosmetic "women" category.

Family with coupled constraints

"Optimize the week, not the brochure."

Friction: two commutes, school calendars, childcare, shade, groceries, medical access, noise and flexible space.

RAMA promise: saved household needs, scenario commute, school-run map, room-fit planning and "what changes at 7:30am?" neighbourhood briefs.

Positioning discipline: these are need states, not stereotypes. RAMA should personalize by explicit constraints and consented behaviour—not infer sensitive attributes from names, nationality, language or browsing patterns.

Design the edge cases into the core.

Person of Determination / older buyer

"Can I move through the whole route?"

Accessibility is chain-dependent: parking → entrance → lift → threshold → rooms → bathroom → amenities.

Dubai Municipality publishes building requirements for people with disabilities, while Dubai's government portal aligns digital access to WCAG 2.1 AA.²⁰²¹

Product: verified measurements, alternative media, captions/transcripts, keyboard tour mode, no-motion tour, support-person sharing.

Sustainability / operating-cost buyer

"What will this building cost and feel like?"

Al Sa'fat Silver is the minimum requirement for new buildings; higher Gold and Platinum levels add performance requirements.²²

Product: rating evidence, cooling arrangement, orientation, glazing/shading notes, utility data when consented, maintenance history and climate-resilience questions.

Off-plan buyer nearing handover

"What changed, and what do I need to do next?"

Marketing, payment, registration, progress, inspection and handover are typically experienced across different documents and contacts.

Product: milestone timeline, payment schedule, source-linked updates, variance log, snagging task list, warranty and service activation.

Arabic-first / multilingual decision maker

"Give me legal and financial clarity in my language."

Translation must preserve defined terms, numbers, dates and source context. Machine translation alone is unsafe for binding or eligibility content.

Product: Arabic + English canonical glossary, reviewed high-risk content, RTL-native components and side-by-side document summaries.

Accessibility metric

% of listings with a measured, end-to-end access path—not number of "accessible" tags.

Translation metric

High-risk content review coverage and terminology consistency—not words translated.

Sustainability metric

% of claims with certificate / bill / method and retrieval date—not green badges.

The trusted way to decide where you belong.

Category

Evidence-led residential marketplace + advisory workflow.

Primary wedge

Curated Dubai inventory with complete Trust Passports and immersive inspections.

Business model

Brokerage / referral revenue where licensed and disclosed; premium seller evidence packs; B2B tour capture; optional owner care. Never sell ranking without explicit "sponsored" labeling.

North-star outcome

Verified decisions completed: a customer reaches a documented next step with all critical evidence reviewed, assumptions acknowledged and no unresolved blocker hidden.

Anti-goal

Maximizing unqualified leads, addictive browsing, pressure countdowns or opaque recommendation fees.

Trust complete

Critical evidence coverage $\geq 90\%$; stale evidence visibly degrades.

Decision useful

Shortlist-to-qualified-viewing conversion; fewer wasted physical visits.

Commercially fair

Revenue source disclosure; organic order separated from sponsorship.

Operationally calm

SLA, advisor handoff and evidence requests live in one record.

Brand promise: "RAMA shows the home, the evidence and the trade-off in the same breath."

Four pillars. One continuous decision record.

01

RAMA Discover

Constraint-first search.

- Life-fit brief
- Evidence coverage
- Map + commute scenarios
- Comparable transactions
- Curated editorial sets

02

RAMA Verify

Property Trust Passport.

- Permit / broker / project
- Document provenance
- Costs + returns
- Risk / unknowns
- Freshness + audit log

03

RAMA Tour

Immersive inspection.

- 360 + 3D + plan
- Measurement + annotations
- Guided live mode
- Light / view scenarios
- Accessible alternatives

04

RAMA Continue

Transaction to ownership.

- Advisor workspace
- Secure documents
- Finance + offer
- Handover / snagging
- Owner care + resale

Sequencing rule

Build Verify before broadening Discover. Build structured tours before XR. Build the decision record before sophisticated AI. Trust debt is harder to repay than technical debt.

MVP supply strategy: 150–300 evidence-complete properties across 6–8 communities; prioritize quality, freshness and learnable operational throughput over catalogue size.

Organize around decisions, not company departments.

RAMA Real-Estate — global: Buy · Rent · Off-plan · Neighbourhoods · Saved · Ask RAMA · Language

Discover	Property	Decide	Continue	Trust
Guided brief	Overview	Compare	Tasks	Methods
Search + map	Trust Passport	Scenario lab	Documents	Sources
Collections	Tour	Questions	Advisor	Corrections
Neighbourhood atlas	Costs / returns	Viewing	Handover	Privacy
Transaction explorer	Building + area	Offer readiness	Ownership	Disclosures

Property page order	URL model	Search model
Fit summary → evidence status → key facts → tour → costs → building / area → risks / unknowns → advisor. Price and CTA never outrank uncertainty.	Stable canonical property / project / building / community IDs; locale segment; state in query; no personally sensitive filters in URL.	Hybrid keyword, facets, geo and intent. Filters explain their basis: measured, source-provided, inferred or unavailable.

Arabic readiness: logical CSS properties, mirrored directional icons only when semantically directional, Arabic line-height tokens, bilingual number / date formatting and a translation memory for regulated terms.

A guided path from uncertainty to readiness.

01 Brief

Household, budget, timeframe, commute, access, tenure and non-negotiables.

02 Readiness

Indicative cash needed, LTV boundaries, programme eligibility and documents.

03 Shortlist

Fit + evidence score; reasons; trade-offs; diverse comparable options.

04 Inspect

Self-guided tour, live expert tour, question capture and physical visit.

05 Decide

Comparable evidence, full cost, offer scenario, unresolved items.

06 Continue

Secure room, finance, offer, registration, handover and owner record.

Critical moments

MOMENT	RAMA RESPONSE
"I can afford the monthly payment."	Reveal cash-to-close, recurring costs and rate sensitivity before property ranking.
"The listing says verified."	Open evidence drawer with source, timestamp, scope and expiry.
"The render looks bigger."	Overlay plan, measurements, camera position and "show unit vs representative" status.
"I need to ask my family."	Share a privacy-safe comparison room with comments and decision criteria.

Recommendation contract

Each recommended property states:

- why it matches;
- which constraints it fails;
- what is verified or assumed;
- whether compensation influenced visibility;
- what the user can change.

A trust signal with an inspectable interior.

Evidence classes

Registry / regulator

DLD-linked or other official source.

Document verified

Reviewed artifact; scope and expiry noted.

On-site observed

Captured by trained operator at a date/time.

Provider attested

Broker / developer claim with accountability.

Modelled

Calculation with method and inputs.

Unverified / unknown

Visible gap; never silently omitted.

RAMA TRUST PASSPORT

82% evidence complete

Residence 1204 · Sample

Permit

QR checked

Availability

Observed 2d ago

Tour

Exact unit

Advert permit + broker

Verified · official

Title / authority to market

Document checked

Service charge

FY2025 evidence

Cooling arrangement

Provider attested

Step-free route

Measured on site

Afternoon construction noise

Unknown · inspect

Scores summarize evidence coverage, not property quality, legal status or investment merit. Open every item to inspect method, artifact, source date and expiry.

Freshness

Time-sensitive signals decay automatically; stale is a state, not a hidden backend problem.

Corrections

Users and providers can challenge facts; updates preserve an audit trail and notify affected shortlists.

No false authority

RAMA verification never substitutes for DLD, legal review, valuation or physical inspection.

Show the whole money story— and the assumptions underneath.

Buyer cash waterfall

LAYER	EXAMPLES
At reservation	Booking / deposit; project-specific payment.
At transaction	Transfer / registration, trustee, agency, valuation, mortgage-related and conveyancing costs as applicable.
During ownership	Mortgage, service charges, cooling / utilities, insurance, maintenance reserve, management.
At exit	Agency, NOC / discharge, settlement, vacancy and sale timing.

Do not hard-code fees into editorial copy. Maintain effective dates, applicability rules, currency and source; require a pre-transaction refresh.

Scenario lab

Purchase price	AED 1.80M
Down payment	Editable
Rate / term	Base · +1% · –1%
Service charge	Evidence dated
Rent / vacancy	Range, not promise
FX	Home-currency view
Exit	Hold 3 / 5 / 10 yrs

Owner-occupier lens

Cash-to-close, monthly carrying cost, rent comparison, emergency reserve and break-even horizon.

Investor lens

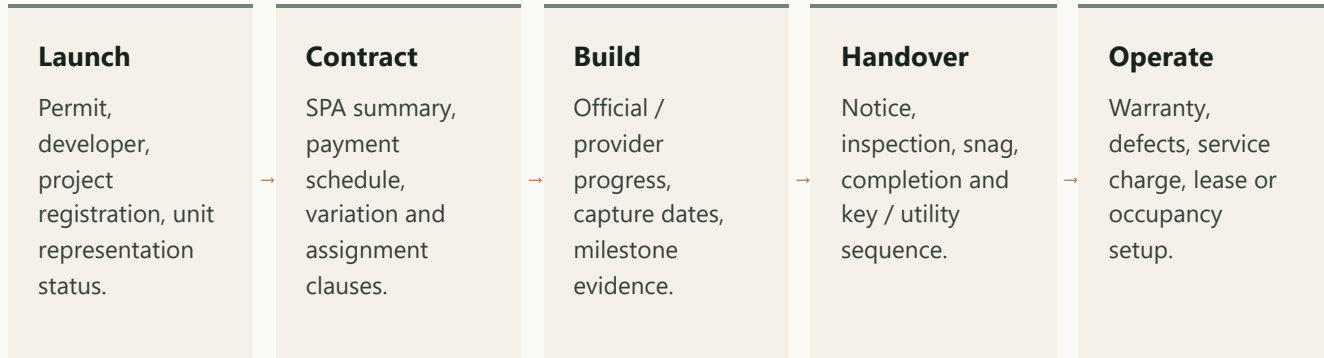
Gross → net yield, vacancy, management, maintenance, financing and resale sensitivity.

Off-plan lens

Payment timeline, financing uncertainty, delay scenarios, handover cash and post-handover ramp.

Guardrail: calculations are deterministic and versioned. AI may explain results, but it must never invent rates, fees, eligibility, projected returns or source dates.

Turn a promise into a monitored chronology.



Project observability model

SIGNAL	DISPLAY RULE
Progress	Show source, capture / publication date and whether percentage is official, developer-provided or RAMA-observed.
Expected date	Separate contractual, current stated and scenario dates.
Show unit / render	Label "exact unit", "same type", "representative finish" or "artist impression" at media level.
Payment	Link each installment to trigger, due-date rule, reminder and evidence.
Change	Preserve what changed, when, by whom and buyer acknowledgement.

Off-plan risk questions

- Is marketing authorized and QR-verifiable?⁸
- Where must buyer funds be deposited?
- What is registered and when?
- What can change under the SPA?
- What happens under delay or cancellation?
- Which features are contractual vs illustrative?
- What cash is due at handover?

Product boundary: RAMA summarizes and tracks; customer counsel interprets binding terms. Store the source clause alongside every high-risk summary and show "reviewed on" metadata.

Tour for understanding, not spectacle.

RAMA Tour is a synchronized set of spatial views, evidence and questions. The user can move from room to room, plan to panorama, daylight to night, and self-guided to live expert without losing context.

Spatial truth

Exact-unit status, capture date, camera position, measurements, plan alignment and any stitched / reconstructed areas.

Inspection layer

Doors, storage, outlets, finishes, appliances, defects, noise sample context, views and access route.

Decision layer

Save a room, ask a question, compare a measurement, request evidence or assign an advisor follow-up.

“Every hotspot should answer a customer question or prove a claim. If it only glows, remove it.”

Core

Responsive 360 tour + floor plan + accessible media transcript.

Enhanced

3D dollhouse, measurements, guided mode, comparison memory.

Context

Balcony view, sun path, building route, neighbourhood arrival.

XR

Progressive enhancement only; never required to access evidence.

Capture once. Publish many trustworthy views.



Media bundle

ASSET	RECOMMENDED FORMAT / USE
Panorama	Equirectangular AVIF/WebP/JPEG tiles; multi-resolution.
Plans	SVG for overlays; source PDF retained; coordinates aligned.
Model	glTF / GLB + Draco / meshopt where appropriate.
Large context	3D Tiles for building / district scale; Cesium supports streaming 3D Tiles. ²³
Alternative	Ordered stills, room descriptions, captions and transcript.

Capture governance

- Written authority to capture and publish.
- Automatic face, plate and personal-artifact review.
- No undisclosed generative “staging” inside evidence media.
- Separate exact-unit evidence from representative marketing.
- Recapture triggers: material change, occupancy change, listing renewal.
- Chain-of-custody metadata for original and derivatives.

Vendor strategy: support a pluggable ingestion contract for Matterport / other providers, but keep RAMA’s evidence schema, annotations and customer interaction data portable.

Progressive immersion for every device and body.

Tier 0 · Essential

Server-rendered overview, compressed still gallery, ordered room list, plan, transcript and evidence drawer. Works on low bandwidth and keyboard.

Tier 1 · Panorama

Lazy-loaded 360 scenes, plan sync, hotspots, saved viewpoint and guided co-browse. Target stable mid-range mobile first.

Tier 2 · Spatial

3D model / dollhouse and measurements. Load only after device / network capability check and explicit user action.

Tier 3 · XR

WebXR via Three.js is an enhancement for compatible devices; maintain non-XR feature parity for evidence.²⁴

Performance budgets

Initial HTML ≤200KB compressed; initial critical media ≤1.5MB; first panorama preview ≤500KB; interaction at 30–60fps; memory guardrails per device class.

Accessibility target

WCAG 2.2 AA as product target: keyboard, visible focus, captions, reduced motion, alternatives, 24px+ touch targets, no orientation lock.²⁵

Tour interface anatomy

ZONE	PURPOSE
Scene	Spatial view; no CTA overlays across key evidence.
Plan rail	Location, orientation and accessible room navigation.
Evidence drawer	Claim, source, capture date, related documents.
Question tray	Timestamp + viewpoint attached to customer question.
Guide mode	Consent-based synchronized navigation with independent pause.

Do not simulate certainty. Sun paths, virtual furniture, view lines and acoustic context must show method, date / season and whether modelled. Avoid “live” labels for cached or illustrative data.

The moat is an evidence-linked property graph.

Core entity	Identity	Relationships	Freshness	Sensitive?
Property / unit	Canonical RAMA ID + external IDs	Building, project, listing, tour	Mixed per field	No / location risk
Listing	Provider + permit + version	Unit, broker, price, availability	Hours / days	Contact data
Evidence	Artifact hash + source + method	Claim(s), verifier, expiry	Explicit	Often restricted
Tour	Capture + published version	Scenes, plan, hotspots, claims	Recapture rule	Possible personal data
Decision	User / household workspace	Criteria, shortlist, scenarios, tasks	Event-sourced changes	Yes
Consent	Subject + purpose + channel	Processing, vendor, withdrawal	Immutable history	Yes

Claim-level provenance

Store each material claim independently: value, unit, source, observed_at, retrieved_at, valid_to, method, confidence, verifier, supersedes.

Bitemporal history

Know both when a fact was true in the market and when RAMA learned it. Essential for price, availability, project status and corrections.

Human override

Corrections require reason, author, evidence and approval; never mutate the audit trail or silently rewrite generated summaries.

Architecture principle: relational core first, graph-shaped APIs where useful. Do not adopt a graph database until query patterns and scale justify operational complexity.

Modular core, event-ready edges.

Experience

Web / PWA

Advisor console

Partner portal

Capture app

Edge + identity

CDN + WAF

BFF / API gateway

OIDC identity + MFA

Consent / feature flags

RAMA application core · modular monolith

Catalogue + search

Evidence + trust

Decision + CRM

Transaction + care

Spatial + intelligence services

Tour runtime

Media pipeline

Cost engine

AI explanation / retrieval

Data + async

PostgreSQL + PostGIS

Search index

Object storage

Queue / event bus /
warehouse

External ecosystem

DLD / approved APIs

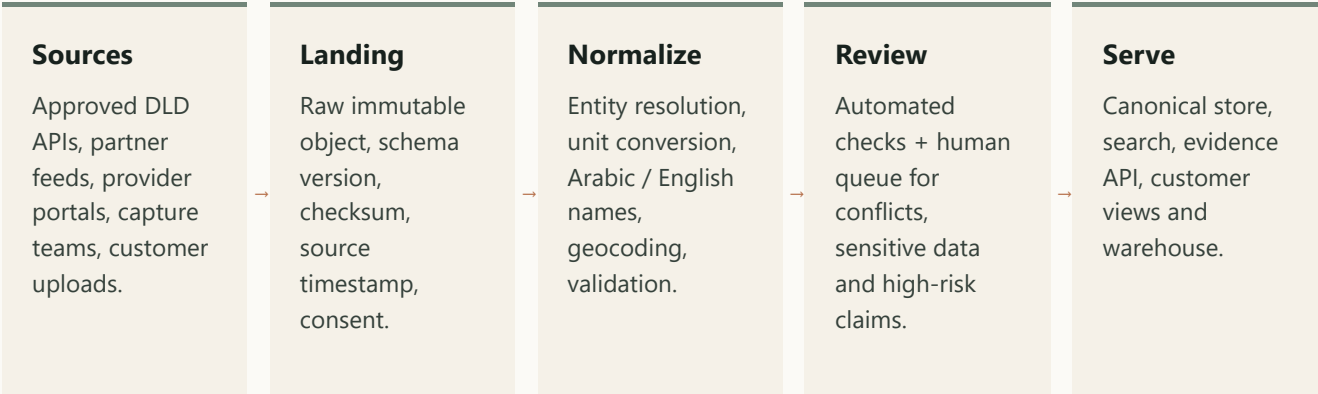
Maps / routing

Mortgage / KYC

CRM / comms / payments

Deploy across multiple Availability Zones in AWS UAE region (me-central-1 lists three AZs).²⁶ Confirm per-service regional availability, data-transfer implications and contractual residency before final design.

Ingest with contracts. Publish with provenance.



Integration principles

Adapter per source — external schemas never leak into the domain.

Idempotent ingestion — replay without duplication.

Reconciliation queues — conflicts become visible work.

Rate + entitlement policy — respect API terms and cache scope.

Field-level lineage — a page can explain where each fact came from.

Kill switch — revoke a source or derivative quickly.

Event backbone

EVENT	CONSUMERS
ListingChanged	Search, freshness, alerts, trust score
EvidenceExpired	Operations queue, page state, notifications
TourPublished	Catalogue, CDN purge, accessibility QA
DecisionUpdated	Advisor, analytics, recommendation context
ConsentWithdrawn	Marketing, personalization, vendors, deletion workflow
OfferStatusChanged	Tasks, document room, customer timeline

API reality: DLD advertises an API gateway and specific services, but availability, commercial terms, permitted storage and downstream display must be confirmed directly before the roadmap depends on them.¹¹

Pragmatic, spatial, UAE-region capable.

LAYER	RECOMMENDATION	WHY / BOUNDARY
Web	Next.js App Router + React + TypeScript	Server rendering, streaming, route architecture, mature ecosystem. App Router uses Server Components, Suspense and Server Functions. ²⁷
UI	CSS variables + accessible headless primitives; Storybook	Own RAMA visual language; avoid library-default aesthetic; test RTL and states.
Core API	NestJS / Fastify, TypeScript; OpenAPI	Modular monolith with explicit domains; shared language with frontend; extract only proven hotspots.
Data	PostgreSQL + PostGIS; Redis	Transactions, geospatial queries, provenance; cache ephemeral state, not truth.
Search	OpenSearch	Facets, geo, Arabic / English analyzers and hybrid retrieval; canonical data remains in Postgres.
Spatial web	Three.js / React Three Fiber; WebXR enhancement	Custom tour interactions and glTF; device-aware runtime. Three.js exposes WebXR management. ²⁴
District 3D	CesiumJS + 3D Tiles when needed	Accurate city-scale geospatial context and streamed LOD; do not use for simple interiors. ²³
Media	S3-compatible object storage + CDN; image / tile workers	Immutable originals, signed derivatives, multi-resolution spatial delivery.
Async	SQS + EventBridge (or equivalent)	Durable jobs, source ingestion, media, notifications and domain events.
Cloud	AWS UAE region, multi-AZ; IaC with Terraform	Regional deployment option and three AZs; confirm service matrix and DR design. ²⁶
Analytics	Consent-aware product events → warehouse + dbt	Decision quality funnels; no raw sensitive text in analytics by default.
Observability	OpenTelemetry + logs, traces, metrics, RUM	Trace customer-visible freshness / tour failures to source and job.

Alternative gate: use a managed 360 / digital-twin vendor to shorten capture launch, but require export rights, stable asset IDs, analytics access, privacy controls and a RAMA-owned annotation layer.

Trust must survive the backend.

Control baseline

Privacy by purpose — consent and lawful-basis mapping; minimize and separate advisory, transaction and marketing data.

UAE PDPL — rights, security, consent exceptions and cross-border controls require a processing register and transfer review.²⁸

Identity — OIDC, MFA for staff / partners, role + relationship-based access, short sessions for document rooms.

Documents — encryption, malware scan, signed URLs, watermarking, download audit, retention policy.

Application — secure SDLC, SAST / DAST / dependency / IaC scanning, secrets manager, WAF and rate limits.

Operations — joiner / mover / leaver, privileged access, immutable audit, incident runbooks and vendor reviews.

Real-estate-specific controls

RISK	CONTROL
Fake / stale ad	Madmoun QR / permit evidence, availability expiry and takedown workflow. ⁸
Broker impersonation	Verified partner identity, registry reconciliation and protected contact channels.
Media privacy	Capture consent; blur review; EXIF removal; no occupied-home personal artifacts.
Recommendation conflict	Compensation disclosure, sponsored separation, model explanation and audit.
Document fraud	Artifact hash, source verification, dual review on high-risk evidence.
AI hallucination	Retrieval-only facts, citations, deterministic calculators, confidence and human escalation.

Retention

Define by artifact and purpose; legal hold separate; delete derivatives and vendor copies through a tracked workflow.

Residency

Document data flows, subprocessors, backups, support access and telemetry—not just primary database region.

Assurance

Threat modelling at design gates; independent penetration test before public transaction / documents launch.

Calm utility. Warm evidence.

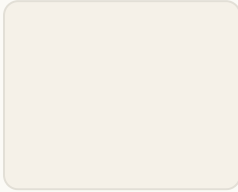
Nordic design is not pale beige minimalism. For RAMA it means democratic access, functional honesty, material restraint, daylight awareness and respect for the user’s time—translated to Dubai’s climate, languages and culture.

Lagom · enough Show the next decision and the evidence it needs. Remove decorative metrics, repetitive CTAs and artificial urgency.	Hygge · reassurance Warm neutral surfaces, human explanations, predictable controls and visible help—without domestic cliché.	Functional honesty Unknowns stay visible; simulated media is labelled; ranking rationale is inspectable.
Democratic design Arabic / English parity, WCAG 2.2 AA target, low-bandwidth paths and no XR requirement.	Nature + climate Use shade, water, stone, native planting and diffused light as editorial signals—not generic Scandinavian snow / pine imagery.	Craft Precise spacing, measured typography, quiet motion, tactile photography and explanatory diagrams.

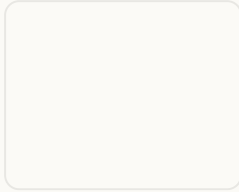
The interface should feel like a good architect at a kitchen table: prepared, precise, unhurried.

Cultural adaptation: avoid importing visual minimalism as cultural neutrality. Validate imagery, privacy, family representations, Arabic typography, prayer / community context and accessibility with Dubai-based users.

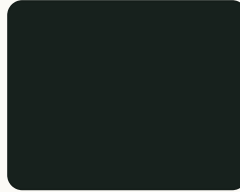
A quiet identity with one warm point of action.



Limestone
#F5F1E8 · canvas



Bone
#FBFAF6 · surface



Ink
#17211D · text



Sage
#718579 · verified



Copper
#B56F49 · action

A home is a long decision.

Editorial serif for decisive, sparse headings. Production choice: a licensed Arabic-compatible family pairing or separate optical equivalents selected together.

RAMA shows what is verified, what is modelled, and what still needs inspection.

Neutral humanist sans for UI, data and long reading. Minimum 16px body on web; comfortable Arabic line height; tabular numerals for financial content.

Spacing + shape

4px base; primary rhythm 8 / 12 / 16 / 24 / 32 / 48 / 64. Corners 6–12px. Use borders and surface changes before shadows.

Motion

120–220ms UI; spatial transitions ≤ 450 ms; meaningful easing; honor reduced motion; never animate urgency.

Iconography

1.5px geometric line, open forms, clear filled alert state. Do not use house-roof clichés for every concept.

Data colour

Colour never acts alone. Verified, review, unknown and risk each include text / icon / pattern.

Make evidence beautiful by making it clear.

Signature components

COMPONENT	REQUIRED ANATOMY
Evidence badge	State, source class, date, open-details action.
Property card	Exact / representative media, price, fit reason, evidence completion, sponsorship.
Cost waterfall	Amount, timing, applicability, assumptions, last reviewed.
Tour hotspot	Question / claim type, evidence link, keyboard target, transcript counterpart.
Risk callout	Observed issue, impact, source, next verification step—no alarmist score alone.
Compare tray	Criteria lock, differences first, missing data visible.

Illustration language

- Subject** — human-scale architecture, thresholds, shade, courtyards, routes and evidence objects.
- Medium** — precise architectural line + soft mineral wash; optional paper grain.
- Palette** — Limestone, Ink and Sage; Copper only for focal evidence / action.
- People** — incidental, diverse, non-tokenizing; privacy-respecting distance.
- Avoid** — skyline spectacle, gold luxury tropes, floating glass UI, neon tech, key clichés.
- AI imagery** — label when illustrative; never blend with property evidence.

Editorial grid

12 columns desktop, 6 tablet, 4 mobile. Max reading line 68 characters. Asymmetry allowed; information alignment remains strict.

Photography

Natural light, corrected verticals, honest lens, consistent white balance. State retouching; retain source original.

Tour UI

Dark spatial chrome only while touring; controls collapse; evidence drawer uses light reading surface; no pulsing hotspots.

Layout test: every page / screen must still communicate hierarchy in grayscale, at 200% zoom, with Arabic copy expansion and with all imagery replaced by descriptive placeholders.

Four phases with evidence-based gates.

PHASE 0 · PROVE 0–8 weeks • Licensing / operating model • DLD / partner API discovery • 30-user research incl. edge groups • Trust schema + content method • Tour capture pilot x10 • Clickable bilingual prototype • Security / privacy DPIA start GATE · Users understand evidence states; viable source access; capture economics.	PHASE 1 · CONCIERGE MVP 2–6 months • 150–300 curated properties • Search / compare / Passport • 360 tour + plan + transcript • Cost engine v1 • Advisor + viewing workflow • Arabic / English core • Analytics + operations QA GATE · ≥90% critical evidence coverage; qualified viewing lift; SLA achieved.	PHASE 2 · MARKETPLACE 6–12 months • Partner portal + feed adapters • Automated freshness / conflicts • Off-plan chronology • Guided live tour • Secure document room • Mortgage referral integration • Community scenario atlas GATE · Partner data quality; trust correction time; repeatable unit economics.	PHASE 3 · SPATIAL LIFECYCLE 12–24 months • 3D / XR selective rollout • Building + district context • Handover / snagging • Owner care + resale • Privacy-safe personalization • Evidence API / B2B product • Selective UAE expansion GATE · Spatial feature utility, lifecycle retention and governance readiness.
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A small team with heavy domain ownership.

Initial product squad · 12–16

CAPABILITY	INDICATIVE OWNERSHIP
Product / domain	Product lead; real-estate operations lead; compliance / counsel fractional.
Design / research	Product designer; design systems / content designer; user researcher fractional.
Engineering	Tech lead; 3–4 full-stack; data / search; spatial engineer; platform / SRE fractional.
Evidence operations	Verification lead; 2 analysts; capture partner network.
Customer	Licensed / qualified advisors based on business model; operations coordinator.

Separate revenue targets from evidence approval. Sales / partnerships cannot override verification state or correction records.

Governance forums

Weekly · evidence quality + stale inventory

Biweekly · product discovery + accessibility

Monthly · privacy / security / AI review

Quarterly · source entitlement + vendor audit

Per release · go / no-go decision with rollback

≥90%

critical evidence coverage on active inventory

<48h

median correction resolution for critical listing facts

–30%

target avoidable physical viewings vs baseline pilot

AA

WCAG 2.2 conformance target for customer journeys

Trust metrics

Evidence open rate; stale rate; correction rate; source coverage; undisclosed conflict incidents.

Decision metrics

Qualified shortlist; useful tour completion; unresolved blocker rate; offer readiness.

Business metrics

Contribution margin per verified decision; partner retention; capture payback; lifecycle revenue.

What could break trust—and how to contain it.

RISK	EARLY INDICATOR	MITIGATION / GATE
Source access assumed	API / display terms remain informal	Prototype with approved manual evidence; no scale promise until signed entitlement.
Verification becomes a vague badge	Users cannot explain states	Usability threshold; open details; claim-level scope; prohibit unlabeled aggregate “verified”.
Stale curated supply	Growing expired evidence queue	Freshness SLAs, automatic degradation, partner scorecards, inventory cap.
Tour cost exceeds value	Low tour use / no viewing reduction	Tier capture by property value and need; measure decision utility before 3D / XR.
Legal / financial summary drift	Outdated fees or eligibility	Effective-dated content, named owner, source monitor, mandatory refresh at action point.
AI creates false confidence	Unsourced answers / hidden assumptions	Retrieval + citations, deterministic calculation, bounded intents, escalation, audit set.
Broker incentives distort ranking	Sponsored stock overperforms without label	Separate ranking lanes; compensation disclosure; independent trust governance.
Nordic aesthetic excludes Dubai	Low Arabic / local trust scores	Co-design in Dubai; bilingual parity; cultural and accessibility research gate.
Microservice overbuild	Slow delivery, distributed failures	Modular monolith; extraction requires independent scale / ownership / reliability case.

Launch readiness is not “features complete.” It is evidence operable, corrections fast, risks disclosed and customer tasks recoverable.

90-day decision: proceed to concierge MVP only if RAMA can secure viable inventory authority, source enough critical evidence, demonstrate that users understand the Passport, and capture a tour that changes a real viewing decision.

Primary sources and public evidence.

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- 2 Dubai Land Department. "Dubai's real estate transactions surge 31% to reach AED 252 billion in Q1 2026." 9 Apr 2026. dubailand.gov.ae/.../q1-2026
- 3 Dubai Land Department. "Dubai's rental sector records strong growth in 2025." 23 Feb 2026. dubailand.gov.ae/.../rental-sector-2025
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- 7 Dubai Land Department. "Dubai Launches Landmark Initiative Further Enabling First-Time Homeownership." 2 Jul 2025. dubailand.gov.ae/.../first-time-home-ownership
- 8 Dubai Land Department. "Madmoun service to verify validity of real estate ads via QR codes." 18 Apr 2023. dubailand.gov.ae/.../madmoun
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17 Nakheel. New launches, app and residential viewing / reservation FAQ. nakheel.com/en/new-launches · nakheel.com/en/contact-us

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22 Dubai Municipality. Al Sa'fat — Dubai Green Building System. dm.gov.ae/.../al-safat-dubai-green-building-system

23 Cesium. Platform and 3D Tiles pipeline. cesium.com/platform · cesium.com/.../3d-tiling-pipeline

24 Three.js. WebXRManager documentation. threejs.org/docs/pages/WebXRManager.html

25 W3C. Web Content Accessibility Guidelines (WCAG) 2.2. w3.org/TR/WCAG22

26 Amazon Web Services. AWS Regions and Availability Zones; UAE me-central-1. docs.aws.amazon.com/.../aws-availability-zones

27 Next.js. App Router documentation. nextjs.org/docs/app

28 UAE Government Portal. Data protection laws; Federal Decree-Law No. 45 of 2021. u.ae/.../data-protection-laws

Accessed 17 July 2026. Public product surfaces change frequently. Re-run the competitive and regulatory audit before roadmap approval. Market claims in this report use the periods stated; no return forecast is implied.